Project: Worked examples
Title: Fillet weld (EN 1993-1-8)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

A fillet weld carrying forces parallel to, transverse to and normal to its axis, by the simplified method: the resultant over the throat area against the design shear strength of the weld, with the required throat and leg.

WHAT IT CALCULATES	RELATION
Resultant force on the weld	$F = \sqrt{(F_{\parallel}^2 + F_{\perp}^2 + F_n^2)}$
Design shear strength of the weld, simplified method	$f_{vw,d} = f_u/(\sqrt{3}\cdot\beta_w\cdot\gamma_{M2})$
Shear stress on the throat	$\tau_w = F/(a\cdot L)$
Weld resistance and utilisation	$F_{w,Rd} = f_{vw,d}\cdot a\cdot L$; $u = F/F_{w,Rd}$
Throat needed (not less than 3 mm) and the matching leg	$a_{req} = \max(F/(f_{vw,d}\cdot L), 3\ \text{mm})$; leg $= \sqrt{2}\cdot a$

Basis: Simplified method for design resistance of fillet weld[1](4.5.3.3); Effective throat thickness: minimum 3 mm[1](4.5.2(2)); Effective length: welds shorter than 30 mm or 6 throats not to carry load[1](4.5.1(2)); Long joints[1](4.11).

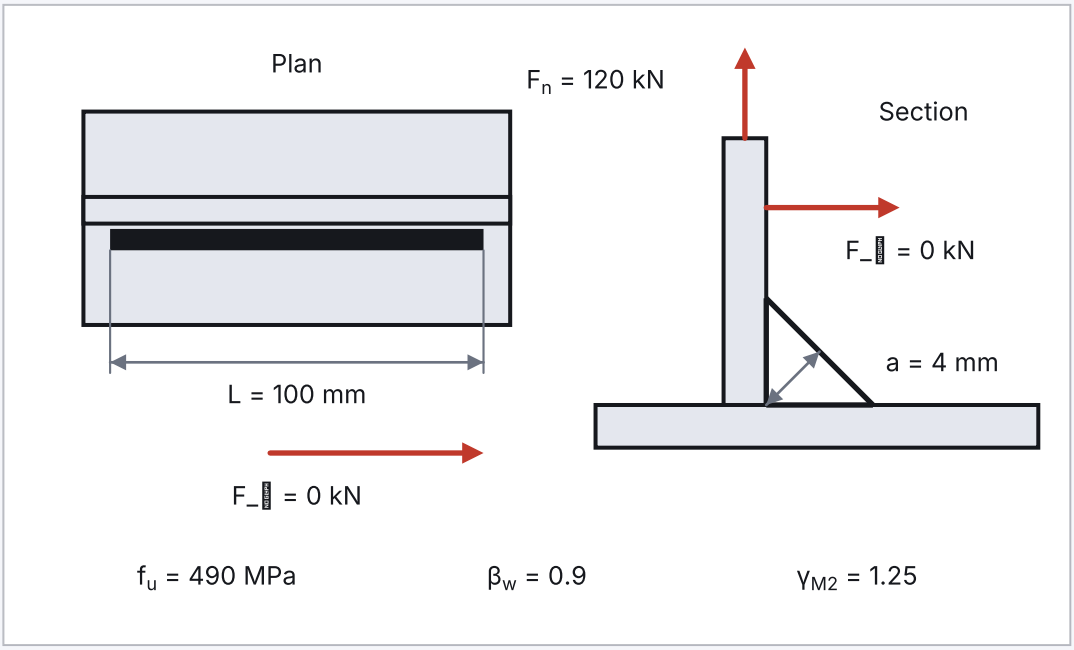


Figure 1 — A fillet weld of effective length L: in plan, the force along it; in section, the throat a and the forces across it and normal to the plate. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
F_{par}	Force parallel to the weld axis	0	kN	spec: LOAD-SCHEDULE-01	-100000 … 100000 kN
F_{tr}	Force transverse to the weld axis, in the plane of the parts	0	kN	spec: LOAD-SCHEDULE-01	-100000 … 100000 kN

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
F_n	Force normal to the plane of the parts	120	kN	spec: LOAD-SCHEDULE-01	-100000 … 100000 kN
L	Effective weld length	100	mm	drawing: DRG-WD-702 rev A	30 … 10000 mm
a	Throat thickness	4	mm	drawing: DRG-WD-702 rev A	3 … 50 mm
f_u	Ultimate strength of the weaker part joined	490	MPa	spec: SPEC-STR-02 design criteria	300 … 1000 MPa
β_w	Correlation factor β_w (0.8 S235, 0.85 S275, 0.9 S355, 1.0 S420/S460)	0.9		spec: SPEC-STR-02 design criteria	0.8 … 1
γ_{M2}	Partial factor γ_{M2}	1.25		spec: SPEC-STR-02 design criteria	1 … 2

3. Assumptions

A1. The resultant force is shared uniformly along the effective length. (Simplified method.)
A2. Equal-leg fillet: leg = √2 × throat. (Geometry.)

4. Working

[9] $F = \sqrt{F_{par}^2 + F_{tr}^2 + F_n^2}$
$= \sqrt{(0\ \text{kN})^2 + (0\ \text{kN})^2 + (120\ \text{kN})^2}$
$= 120\ \text{kN}$
Resultant force on the weld

[10] $f_{vw,d} = f_u / (\sqrt{3} * \beta_w * \gamma_{M2})$ [1](4.5.3.3)
$= 490\ \text{MPa} / (\sqrt{3} \times 0.9 \times 1.25)$
$= 251.47\ \text{MPa}$
Design shear strength of the weld

[11] $\tau_w = F / (a * L)$
$= 120\ \text{kN} / (4\ \text{mm} \times 100\ \text{mm})$
$= 300\ \text{MPa}$
Stress on the throat

[12] $F_{w,Rd} = f_{vw,d} * a * L$ [1](4.5.3.3)
$= 251.47\ \text{MPa} \times 4\ \text{mm} \times 100\ \text{mm}$
$= 100.59\ \text{kN}$
Design resistance of the weld

[13] $u = F / F_{w,Rd}$
$= 120\ \text{kN} / 100.59\ \text{kN}$
$= 1.193$
Utilisation

[14] $a_{calc} = F / (f_{vw,d} * L)$
$= 120\ \text{kN} / (251.47\ \text{MPa} \times 100\ \text{mm})$
$= 4.772\ \text{mm}$
Throat needed for the load

[15] $a_{req} = \max(a_{calc}, 3\ \text{mm})$ [1](4.5.2(2))
$= \max(4.772\ \text{mm}, 3\ \text{mm})$
$= 4.772\ \text{mm}$
Required throat (not less than 3 mm)

[16] $s_{leg} = \sqrt{2} * a$
$= \sqrt{2} \times 4\ \text{mm}$
$= 5.6569\ \text{mm}$

Leg length of the throat provided (equal legs)

5. Results

SYMBOL	VALUE	DESCRIPTION
F	120 kN	Resultant force
$f_{vw,d}$	251.5 MPa	Design shear strength
τ_w	300 MPa	Throat stress
$F_{w,Rd}$	100.59 kN	Weld resistance
a_{calc}	4.772 mm	Throat needed
a_{req}	4.772 mm	Required throat
s_{leg}	5.657 mm	Leg length provided
u	1.20	Utilisation (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Weld resistance	u = 1.20	u ≤ 1	1.20	[1](4.5.3.3)

7. Applicability

None.

8. Verification

Template struct.fillet-weld@1.0.2 Content hash 28bde48078cccba1 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-8: Design of Joints. BS EN 1993-1-8:2005. BSI; 2005.